

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$\underline{x}^2 + 8x + \underline{16} = (x)^2 + 2 \times x \times 4 + 4^2$$

$$= (x+4)^2$$

$$9\underline{x}^2 - 30x + 2\underline{5} = (3x)^2 - 2 \times 3x \times 5 + (5)^2$$

$$\underline{x}^2 + \underline{x} + \frac{1}{4} = (3x-5)^2$$

$$(x)^2 + 2 \times \textcircled{x} \times \frac{1}{2} + \left(\frac{1}{2}\right)^2$$

$$p^4 - (p+q)^4$$

$$\begin{aligned} p^4 - (p+q)^4 &= p^2 - 2 \times p^2 \times (p+q)^2 + (p+q)^2 \\ &= (p^2 - (p+q)^2)^2 \end{aligned}$$

$$\overbrace{(x+2)(x-2)} \quad \boxed{(x^2-4)}$$

$$(a+b)(a-b) = a^2 - b^2$$

$$(x+2)(x-2) = x^2 - 2^2$$

$$= \underline{\underline{x^2 - 4}}$$

$$(x^2 - 4)(x^2 - 4)$$

$$(x^2 - 4)^2 = (a - b)^2 = a^2 - 2ab + b^2$$

$$= (x^2)^2 - 2 \times x^2 \times 4 + 4^2$$

$$= \underline{\underline{x^4 - 8x^2 + 16}}$$

If marked price = ₹ 990 and percentage of discount is 10, then find the selling price.

$$MP = ₹ 990$$

$$D = 10\%$$

$$\text{Discount} = \frac{D\%}{100} \times MP$$

$$= \frac{10}{100} \times 990$$

$$\begin{aligned} SP &= ₹ 99 \\ &= MP - D \\ &= 990 - 99 \\ &= ₹ 891 \end{aligned}$$

The marked price of a mixer is 2300 rupees. A customer purchased it for Rs.1955. Find percentage of discount offered to the customer.

$$\begin{aligned}MP &= ₹ 2300 \\SP &= ₹ 1955 \\Discount &= MP - SP \\&= 2300 - 1955 \\&= ₹ 345\end{aligned}$$

$$\frac{D\%}{100}$$

$$D\%$$

$$\begin{aligned}&= \frac{345}{2300} \times 100 \\&= 15\%\end{aligned}$$

$$\begin{array}{r}34 \\-23 \\ \hline 11 \\ 115 \\ \hline 23 \overline{) 345}\end{array}$$

If selling price = ₹ 900. Discount is 20 %, then find the marked price.

$$\begin{aligned}\text{Let } MP &= x \\ D\% &= 20 \\ D &= \frac{D\% \text{ of } MP}{100}\end{aligned}$$

$$D = \frac{20}{100} \times x$$

$$D = \frac{x}{5}$$

$$SP = MP - \text{Discount}$$

$$900 = \frac{20}{100} \times \frac{x}{5}$$

$$900 = \frac{5x - x}{5}$$

$$900 = \frac{4x}{5} \quad 225$$

$$x = \frac{900 \times 5}{4}$$

$$= 225 \times 5$$

$$= 1125$$

$$\text{Let } (4x + 2y)(x + y)$$

$$4x(x + y) + 2y(x + y)$$

$$MP = 100$$

$$SP = 80$$

$$MP = x$$

$$SP = 900$$

$$\begin{array}{r} 225 \quad 5 \\ \hline \cancel{900} \times \cancel{100} \\ \hline \cancel{80} \\ \cancel{4} \end{array}$$

$$\frac{100}{80} \neq \frac{x}{900}$$

$$x \times 80 = 900 \times 100$$

$$\begin{array}{r} 225 \quad 5 \\ x = \cancel{900} \times \cancel{100} \\ \hline \quad \quad \quad 80 \\ \quad \quad \quad \cancel{4} \\ \hline = 1125 \end{array}$$

$$\frac{\tan A}{(1 + \tan^2 A)^2} + \frac{\cot A}{(1 + \cot^2 A)^2} = \sin A \cos A$$

$$LHS = \frac{\tan A}{(1 + \tan^2 A)^2} + \frac{\cot A}{(1 + \cot^2 A)^2}$$

$$= \frac{\tan A}{(\sec^2 A)^2} + \frac{\cot A}{(\csc^2 A)^2}$$

$$= \tan A \times \cos^4 A + \cot A \times \sin^4 A$$

$$= \frac{\sin A}{\cancel{\cos A}} \times \cos^3 A + \frac{\cos A}{\cancel{\sin A}} \times \sin^3 A$$

$$\begin{array}{r} 1.25 \\ + 3.00 \\ \hline \end{array}$$

$$\begin{array}{r} \sin A \cos^3 A \\ + \cos A \sin^3 A \\ \hline \end{array}$$

$$\boxed{\sin A \cos A} \times (\cos^2 A + \sin^2 A)$$

$$\begin{array}{r} 3.00 \\ - 1.25 \\ \hline \end{array}$$

$$\sec^4 A (1 - \sin^4 A) - 2 \tan^2 A = 1$$

$$\frac{1^2 - \sin^2 A}{\cos^2 A} = \frac{(\sin^2 A)^2}{\cos^2 A}$$

$$\text{L.H.S} = \sec^4 A (1 - \sin^4 A) - 2 \tan^2 A$$

$$= \frac{1}{\cos^4 A} (1 + \sin^2 A)(1 - \sin^2 A) - 2 \tan^2 A$$

$$= \frac{1}{\cos^4 A} \times \cancel{\cos^2 A} \times (1 + \sin^2 A) - 2 \tan^2 A$$

$$= \frac{1 + \sin^2 A}{\cos^2 A} - \frac{2 \sin^2 A}{\cos^2 A}$$

$$= \frac{1 + \sin^2 A - 2 \sin^2 A}{\cos^2 A}$$

$$= \frac{1 - \sin^2 A}{\cos^2 A}$$

$$= \frac{\cancel{\cos^2 A}}{\cancel{\cos^2 A}}$$

$$= 1$$

$$= \text{R.H.S}$$